23MIS0109
VC SAI TEJASWI

1. Simple Java Console App (Baseline CI/CD)
TASK : Create Maven project with Junit tests
Create a Maven based application by following standard Maven directory structure.
🕐 Write the code in Visual Studio and push it to your GITHUB repository.
🕐 Modify the code and commit the updates back in GITHUB.
🕐 Using Jenkins configure the job as pipeline and create the stages for the Continuous Integration process.
🕐 Containerize the application using Dockerfile and display the stages in stage view in Jenkins.

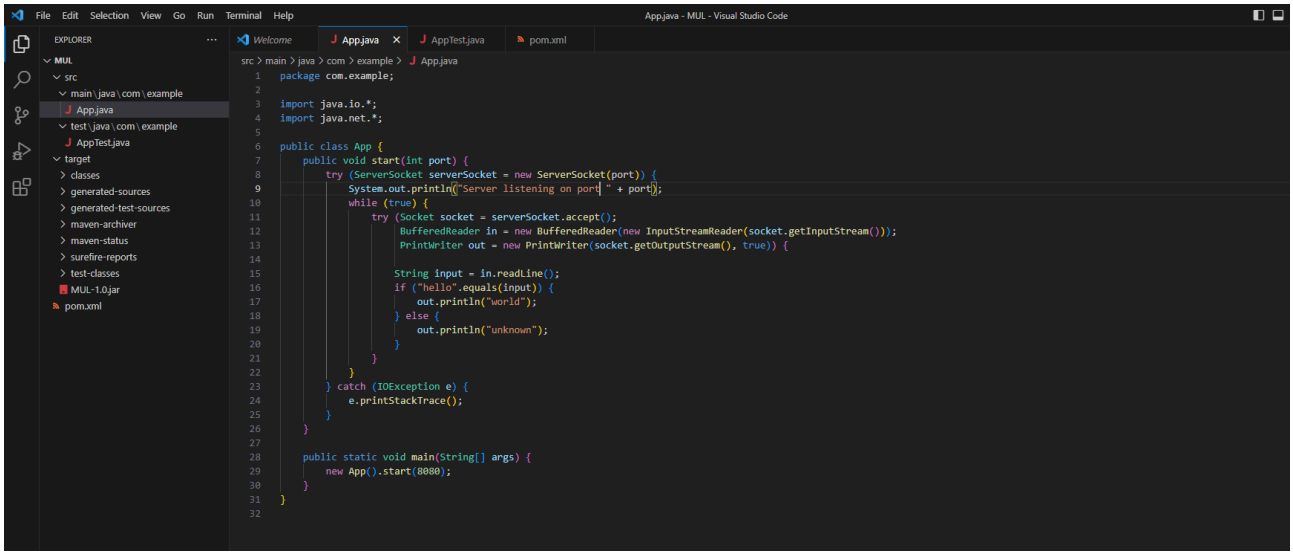
Socket Program

filename is MUL

gitlink:

https://github.com/SAITEJASWI2323/mul

App. Java



AppTest.java:

```java
package com.example;

import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.assertEquals;
import java.io.*;
import java.net.*;

public class AppTest {
    @Test
    public void testServerResponse() throws IOException, InterruptedException {
        // Start server in a background thread for testing
        Thread serverThread = new Thread(() -> new App().start(8888));
        serverThread.setDaemon(true);
        serverThread.start();

        Thread.sleep(1000); // Wait for server to bind

        try (Socket socket = new Socket("localhost", 8888);
             PrintWriter out = new PrintWriter(socket.getOutputStream(), true);
             BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()))) {

            out.println("hello");
            assertEquals("world", in.readLine());
        }
    }
}
```

pom.xml

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.example</groupId>
    <artifactId>MUL</artifactId>
    <version>1.0</version>

    <properties>
        <maven.compiler.source>1.8</maven.compiler.source>
        <maven.compiler.target>1.8</maven.compiler.target>
    </properties>

    <dependencies>
        <!-- Correct JUnit 5 Dependency for your AppTest.java -->
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter-api</artifactId>
            <version>5.8.2</version>
            <scope>test</scope>
        </dependency>
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter-engine</artifactId>
            <version>5.8.2</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

    <build>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-jar-plugin</artifactId>
                <version>3.2.0</version>
                <configuration>
                    <archive>
                        <manifest>
                            <mainClass>com.example.App</mainClass>
                        </manifest>
                    </archive>
                </configuration>
            </plugin>
        </plugins>
    </build>
```

```
create mode 100644 target/surefire-reports/com.example.AppTest.txt
create mode 100644 target/test-classes/com/example/AppTest.class
PS C:\Users\batch1\Desktop\MUL> git branch -M main
```

mvn clean package: build is sucess

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\Users\batch1\Desktop\MUL> mvn clean package
>>
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< com.example:MUL >-------------------------
[INFO] Building MUL 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ MUL ---
[INFO] Deleting C:\Users\batch1\Desktop\MUL\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ MUL ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[WARNING] bootstrap class path not set in conjunction with -source 8
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ MUL ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\batch1\Desktop\MUL\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ MUL ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ MUL ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
Server listening on port 8888
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.053 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.2.0:jar (default-jar) @ MUL ---
[INFO] Building jar: C:\Users\batch1\Desktop\MUL\target\MUL-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.041 s
[INFO] Finished at: 2026-04-09T16:32:41+05:30
[INFO] ------------------------------------------------------------------------
```

Do you want to install the recommended 'Docker' extension from Microsoft for Docker language?

Install Show Recommendations

File Edit Selection View Go Run Terminal Help pom.xml - MUL - Visual Studio Code

EXPLORER

- MUL
  - src
    - main\java\com\example
      - App.java
    - test\java\com\example
      - AppTest.java
  - target
    - classes
    - generated-sources
    - generated-test-sources
    - maven-archiver
    - maven-status
    - surefire-reports
    - test-classes
    - MUL-1.0.jar
  - pom.xml

Welcome | App.java | AppTest.java | pom.xml

pom.xml

```
        </dependency>
    </dependencies>

    <build>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-jar-plugin</artifactId>
                <version>3.2.0</version>
                <configuration>
                    <archive>
                        <manifest>
                            <mainClass>com.example.App</mainClass>
                        </manifest>
                    </archive>
                </configuration>
            </plugin>
        </plugins>
    </build>
```

OUTLINE

TIMELINE

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
[WARNING] bootstrap class path not set in conjunction with -source 8
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ MUL ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
Server listening on port 8888
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.053 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.2.0:jar (default-jar) @ MUL ---
[INFO] Building jar: C:\Users\batch1\Desktop\MUL\target\MUL-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.041 s
[INFO] Finished at: 2026-04-09T16:32:41+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\batch1\Desktop\MUL>
```

pushed the files to git

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL                                                    powershell

[INFO] ------------------------------------------------------------------------
PS C:\Users\batch1\Desktop\MUL> git init
Initialized empty Git repository in C:/Users/batch1/Desktop/MUL/.git/
PS C:\Users\batch1\Desktop\MUL> git add .
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/surefire-reports/TEST-com.example.AppTest.xml', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\batch1\Desktop\MUL> git commit -m "first commit"
[master (root-commit) b4c6fc9] first commit
 Committer: Batch1 <Batch1@VITUNIVERSITY.LOCAL>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
    git commit --amend --reset-author

 13 files changed, 177 insertions(+)
 create mode 100644 pom.xml
 create mode 100644 src/main/java/com/example/App.java
 create mode 100644 src/test/java/com/example/AppTest.java
 create mode 100644 target/MUL-1.0.jar
 create mode 100644 target/classes/com/example/App.class
 create mode 100644 target/maven-archiver/pom.properties
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
 create mode 100644 target/surefire-reports/TEST-com.example.AppTest.xml
 create mode 100644 target/surefire-reports/com.example.AppTest.txt
 create mode 100644 target/test-classes/com/example/AppTest.class
PS C:\Users\batch1\Desktop\MUL> git branch -M main
PS C:\Users\batch1\Desktop\MUL> git remote add origin https://github.com/SAITEJASWI2323/mul.git
PS C:\Users\batch1\Desktop\MUL> git push -u origin main
Enumerating objects: 39, done.
Counting objects: 100% (39/39), done.
Delta compression using up to 20 threads
Compressing objects: 100% (18/18), done.
Writing objects: 100% (39/39), 9.79 KiB | 1.40 MiB/s, done.
Total 39 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/SAITEJASWI2323/mul.git
```

update the App.java code and push to git

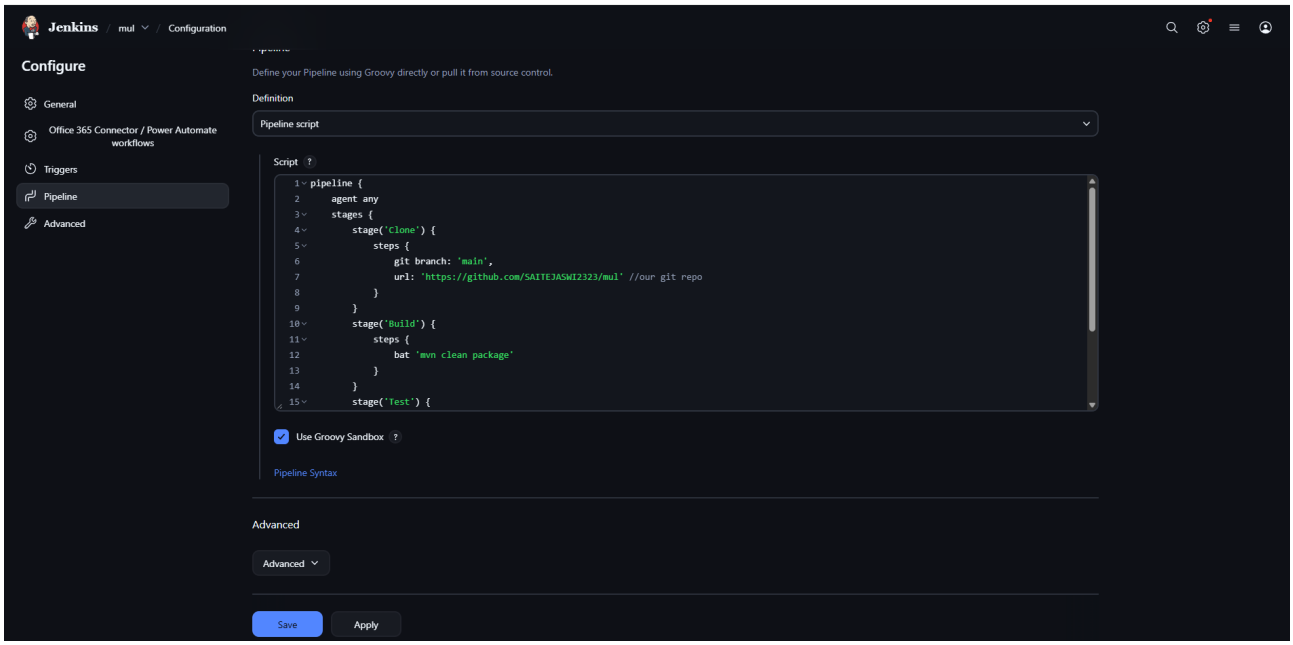
```
File  Edit  Selection  View  Go  Run  Terminal  Help                    App.java - MUL - Visual Studio Code

EXPLORER                    Welcome   App.java M   AppTest.java   pom.xml
MUL                         src > main > java > com > example > App.java
  src
    main\java\com\example
      App.java                M
    test\java\com\example
      AppTest.java
  target
    classes
    generated-sources
    generated-test-sources
    maven-archiver
    maven-status
    surefire-reports
    test-classes
    MUL-1.0.jar
  pom.xml

1  package com.example;
2
3  import java.io.*;
4  import java.net.*;
5
6  public class App {
7      public void start(int port) {
8          try (ServerSocket serverSocket = new ServerSocket(port)) {
9              // UPDATED LINE BELOW
10             System.out.println("Socket Server is now running and listening on port: " + port);
11
12             while (true) {
13                 try (Socket socket = serverSocket.accept();
14                      BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()));
15                      PrintWriter out = new PrintWriter(socket.getOutputStream(), true)) {
16
17                     String input = in.readLine();
18                     if ("hello".equals(input)) {
19                         out.println("world");
20                     } else {
21                         out.println("unknown");
22                     }
23                 }
24             }
25         } catch (IOException e) {
26             e.printStackTrace();
27         }
28     }
29
30     public static void main(String[] args) {
31         new App().start(8080);
32     }
33 }
34
```

pushed updated code to git

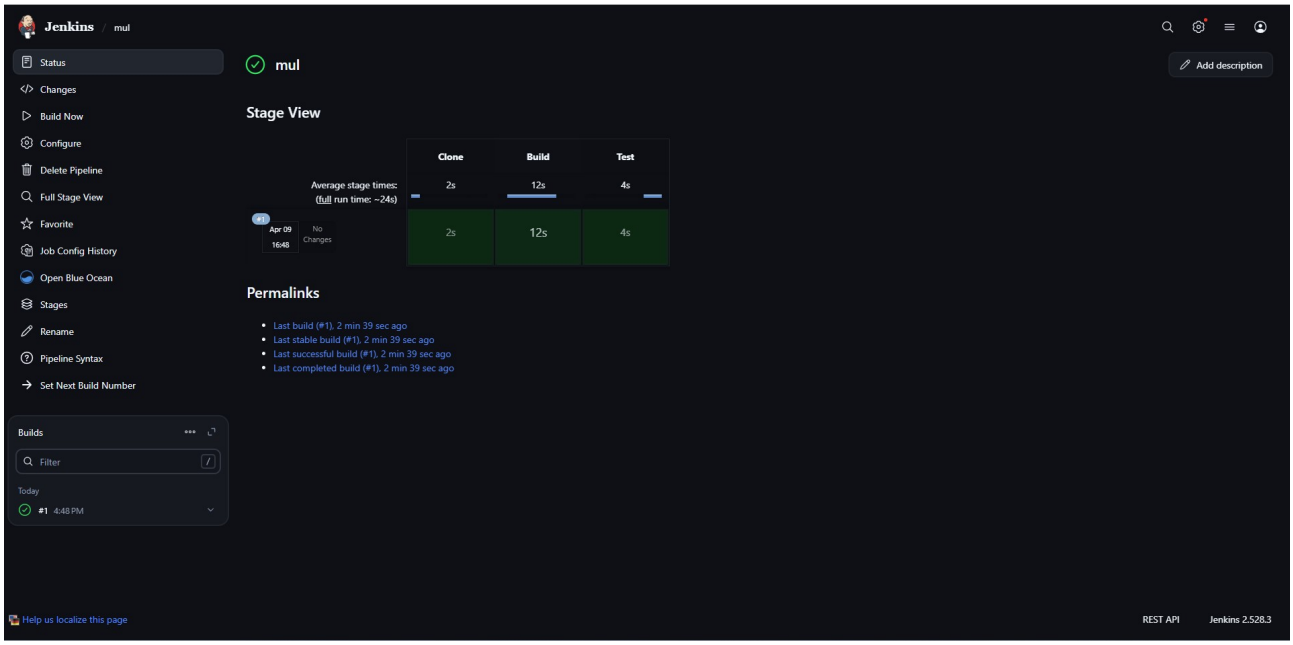
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL                                                    powershell

branch 'main' set up to track 'origin/main'.
Everything up-to-date
PS C:\Users\batch1\Desktop\MUL> git add .
PS C:\Users\batch1\Desktop\MUL> git commit -m "t comt"
[main fde89b7] t comt
 Committer: Batch1 <Batch1@VITUNIVERSITY.LOCAL>
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 3 insertions(+), 1 deletion(-)
PS C:\Users\batch1\Desktop\MUL> git remote add origin https://github.com/SAITEJASWI2323/mul.git
error: remote origin already exists.
PS C:\Users\batch1\Desktop\MUL> git push -u origin main
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 20 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (8/8), 593 bytes | 197.00 KiB/s, done.
Total 8 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/SAITEJASWI2323/mul.git
   54dca49..fde89b7  main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\batch1\Desktop\MUL>
```

create a pipeline in jenkins:

```
pipeline {
    agent any
    stages {
        stage('Clone') {
            steps {
                git branch: 'main',
                url: 'https://github.com/SAITEJASWI2323/mul'
            }
        }
        stage('Build') {
            steps {
                bat 'mvn clean package'
            }
        }
        stage('Test') {
            steps {
                bat 'mvn test'
            }
        }
    }
}
```



build sucesss:



mul #1 Console - Jenkins | SAITEJASWI2323/mul | VIT Vellore - VTOP | VL2025260502387_AST05.pdf

localhost:8085/job/mul/1/console

Jenkins / mul / #1 / Console Output

- Status
- Changes
- Console Output
- Edit Build Information
- Delete build '#1'
- Timings
- Git Build Data
- Rebuild
- Open Blue Ocean
- Pipeline Overview
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces

Console Output

Download | Copy | View as plain text

```
Started by user admin
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\mul
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/SAITEJASWI2323/mul
 > git.exe init C:\ProgramData\Jenkins\.jenkins\workspace\mul # timeout=10
Fetching upstream changes from https://github.com/SAITEJASWI2323/mul
 > git.exe --version # timeout=10
 > git --version # 'git version 2.52.0.windows.1'
 > git.exe fetch --tags --force --progress -- https://github.com/SAITEJASWI2323/mul +refs/heads/*:refs/remotes/origin/* # timeout=10
 > git.exe config remote.origin.url https://github.com/SAITEJASWI2323/mul # timeout=10
 > git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
 > git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision fde89b72efd946fdf63f3bae2ba205aa25fb5ecb (refs/remotes/origin/main)
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f fde89b72efd946fdf63f3bae2ba205aa25fb5ecb # timeout=10
 > git.exe branch -a -v --no-abbrev # timeout=10
 > git.exe checkout -b main fde89b72efd946fdf63f3bae2ba205aa25fb5ecb # timeout=10
Commit message: "t comt"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] bat
```

mul #1 Console - Jenkins | SAITEJASWI2323/mul | VIT Vellore - VTOP | VL2025260502387_AST05.pdf

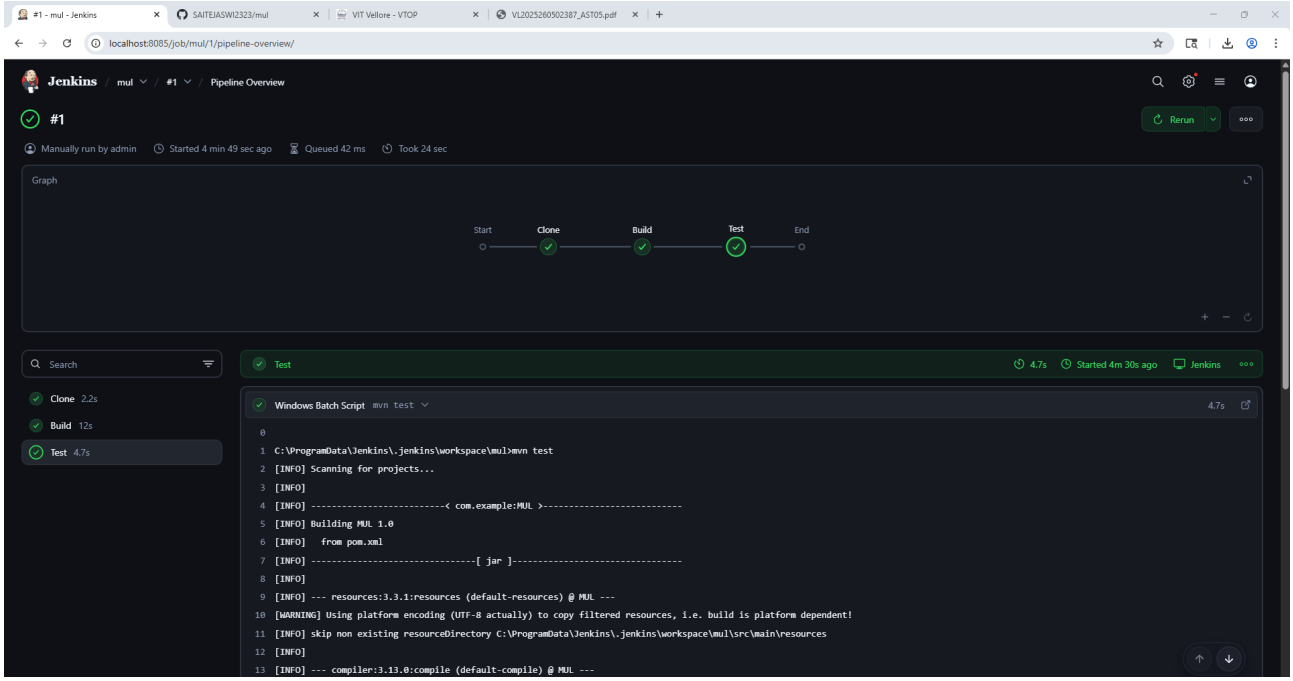
localhost:8085/job/mul/1/console

Jenkins / mul / #1 / Console Output

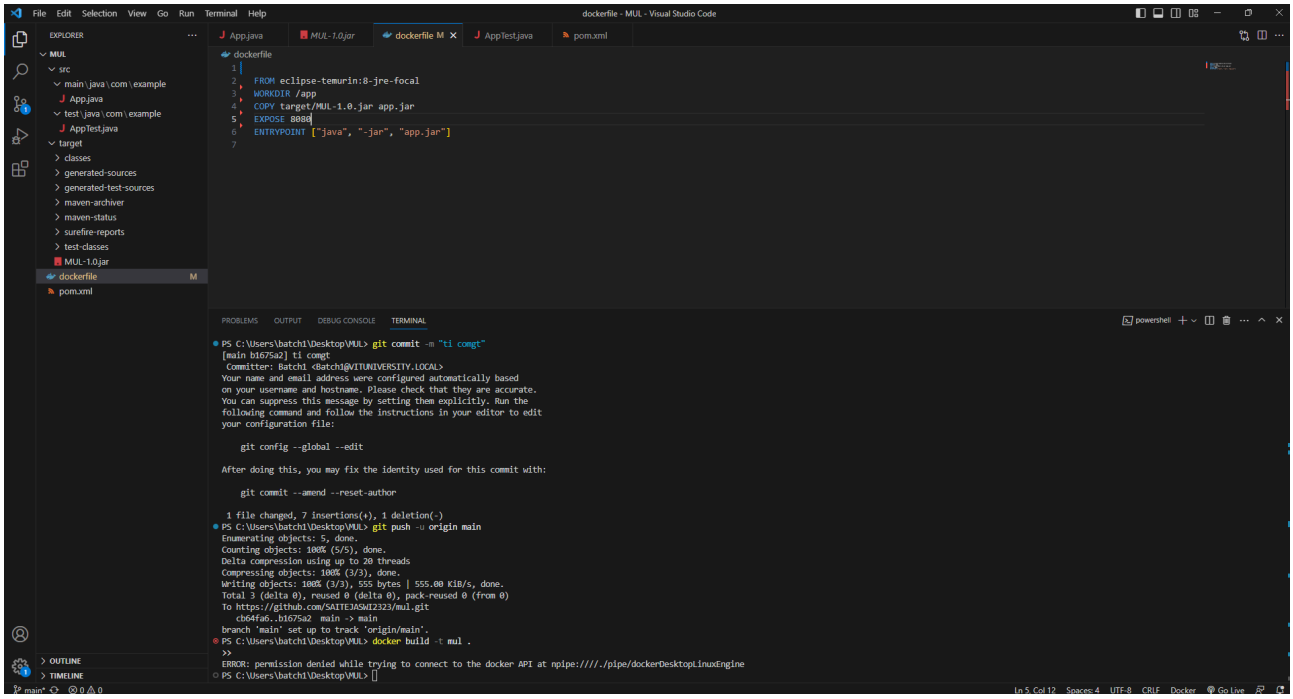
```
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\ProgramData\Jenkins\.jenkins\workspace\mul\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ MUL ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ MUL ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
Socket Server is now running and listening on ports: 8888
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.053 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.086 s
[INFO] Finished at: 2026-04-09T16:48:59+05:30
[INFO] ------------------------------------------------------------------------
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Help us localize this page

REST API | Jenkins 2.528.3



Create Dockerfile: (and push to git)



FROM openjdk:8-jre-slim

WORKDIR /app

COPY target/MUL-1.0.jar app.jar

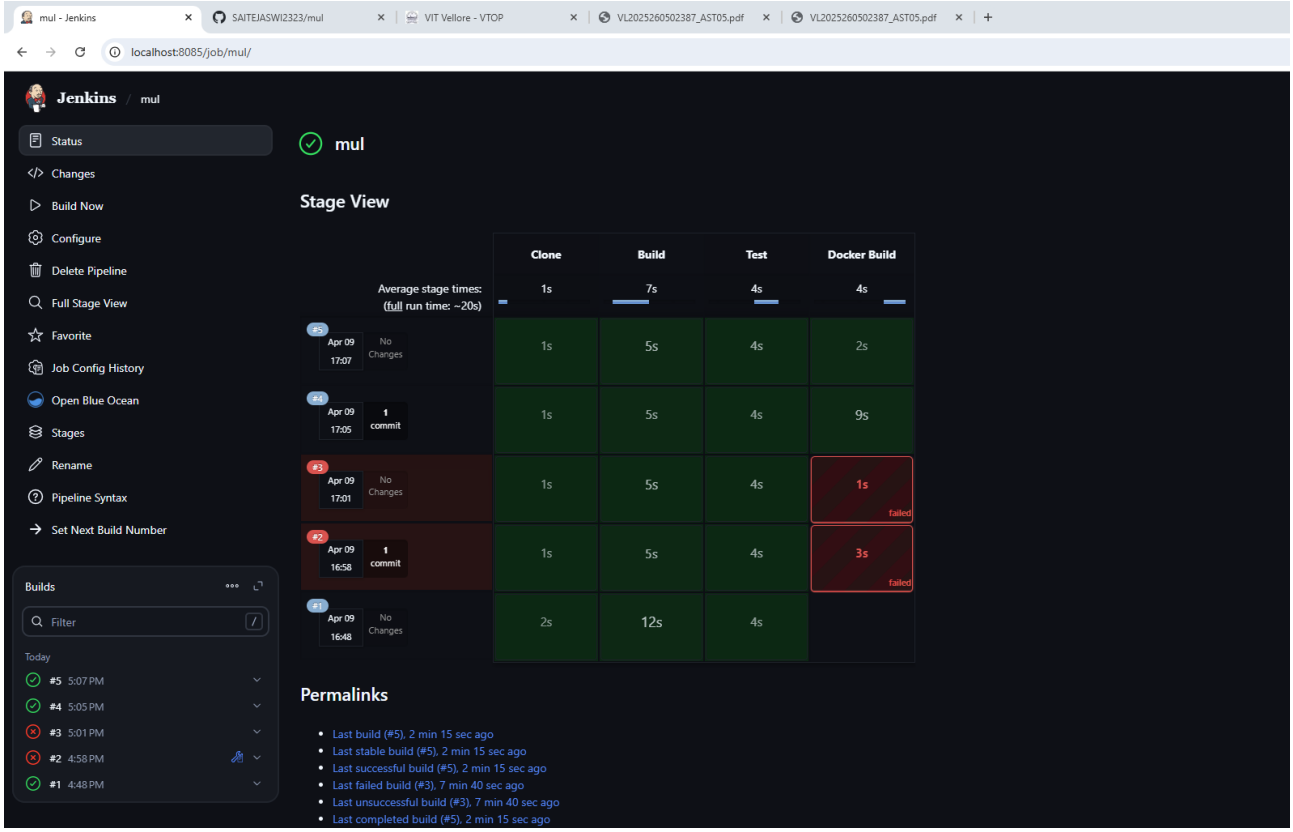
EXPOSE 8080

ENTRYPOINT ["java", "-jar", "app.jar"]

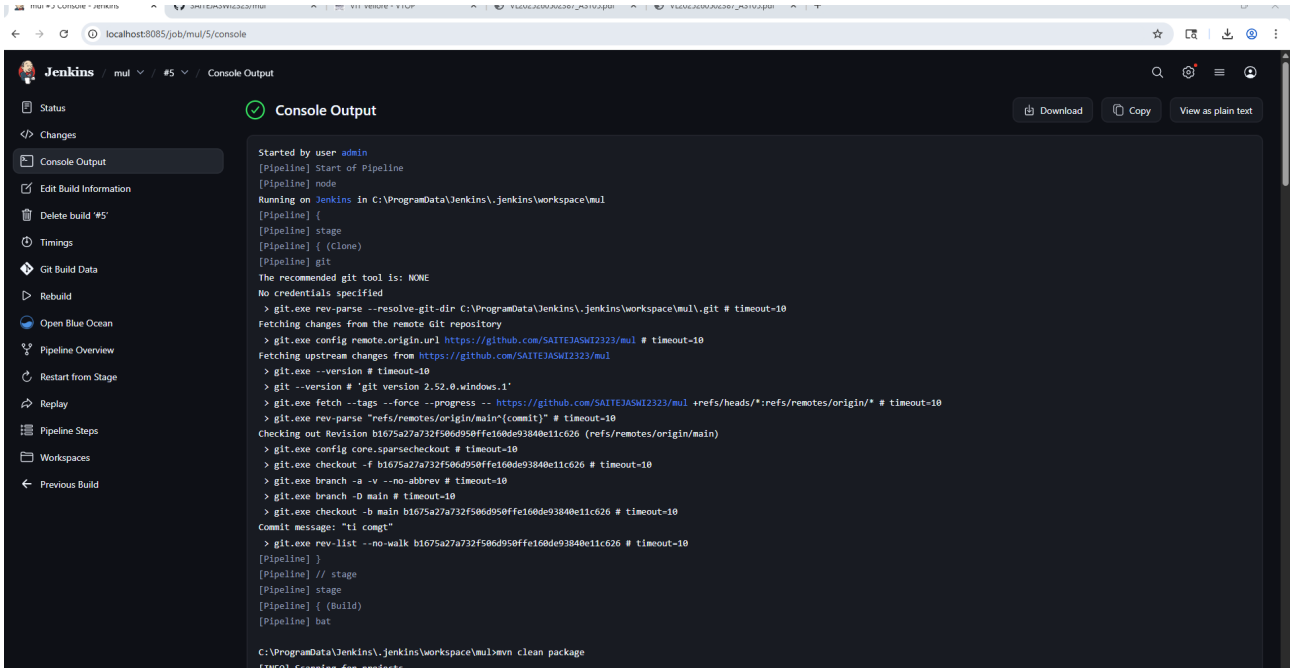
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL                                                    powershell

PS C:\Users\batch1\Desktop\MUL> git add .
PS C:\Users\batch1\Desktop\MUL> git commit -m "ti comt"
[main cb64fa6] ti comt
 Committer: Batch1 <Batch1@VITUNIVERSITY.LOCAL>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 5 insertions(+)
 create mode 100644 dockerfile
PS C:\Users\batch1\Desktop\MUL> git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 451 bytes | 451.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/SAITEJASWI2323/mul.git
   fde89b7..cb64fa6  main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\batch1\Desktop\MUL>
```

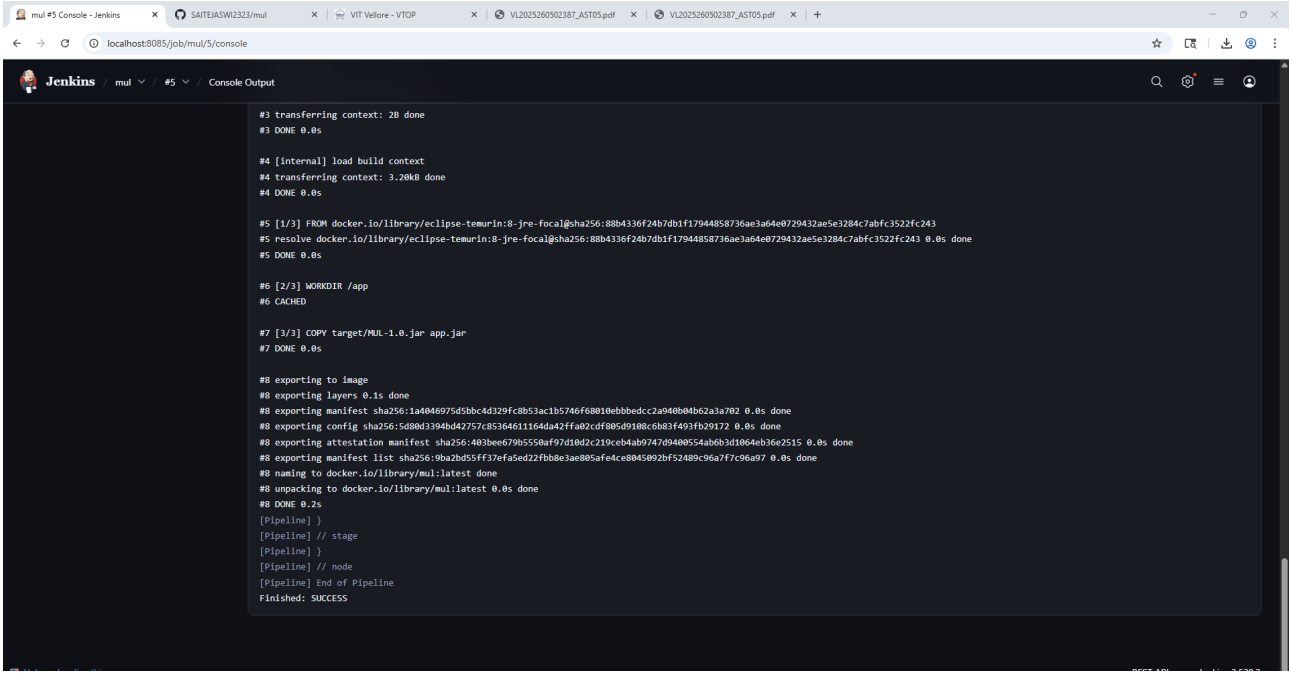
updated pipeline script:

```
pipeline {
    agent any
    stages {
        stage('Clone') {
            steps {
                git branch: 'main',
                    url: 'https://github.com/SAITEJASWI2323/mul' //git repo
            }
        }
        stage('Build') {
            steps {
                bat 'mvn clean package'
            }
        }
        stage('Test') {
            steps {
                bat 'mvn test'
            }
        }
        stage('Docker Build') {
            steps {
                bat 'docker build -t mul .'
            }
        }
    }
}
```



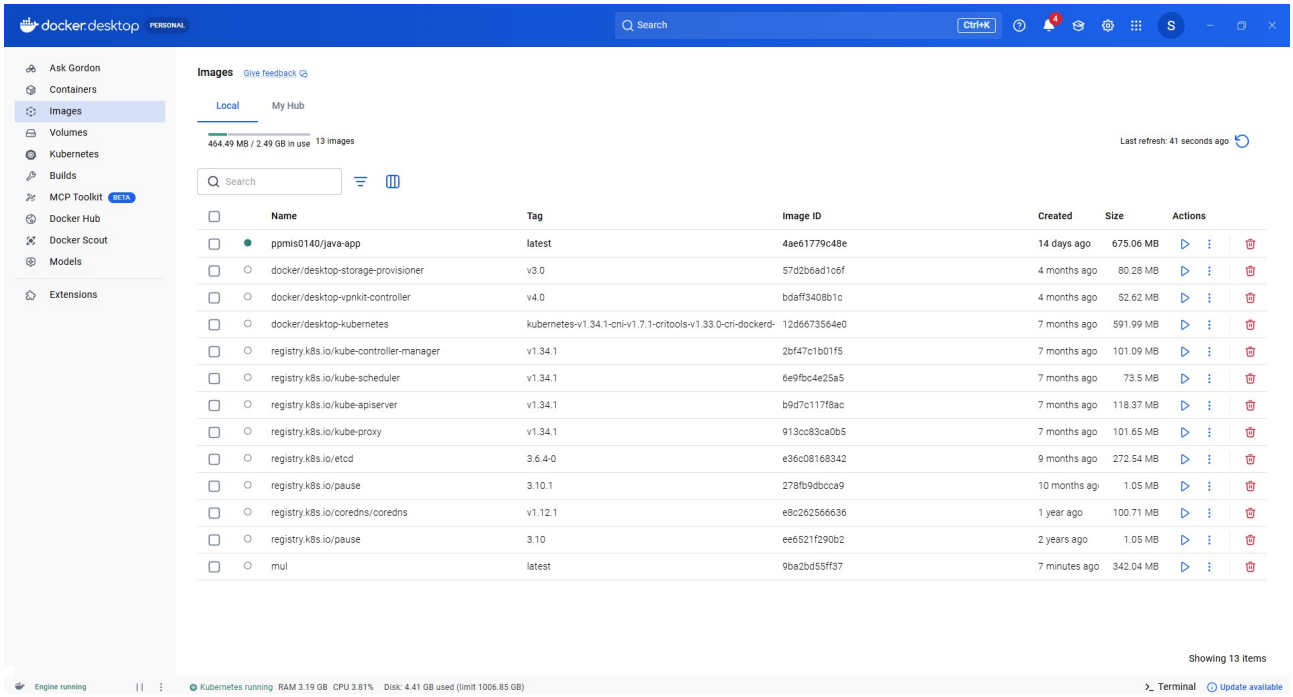
build is sucess



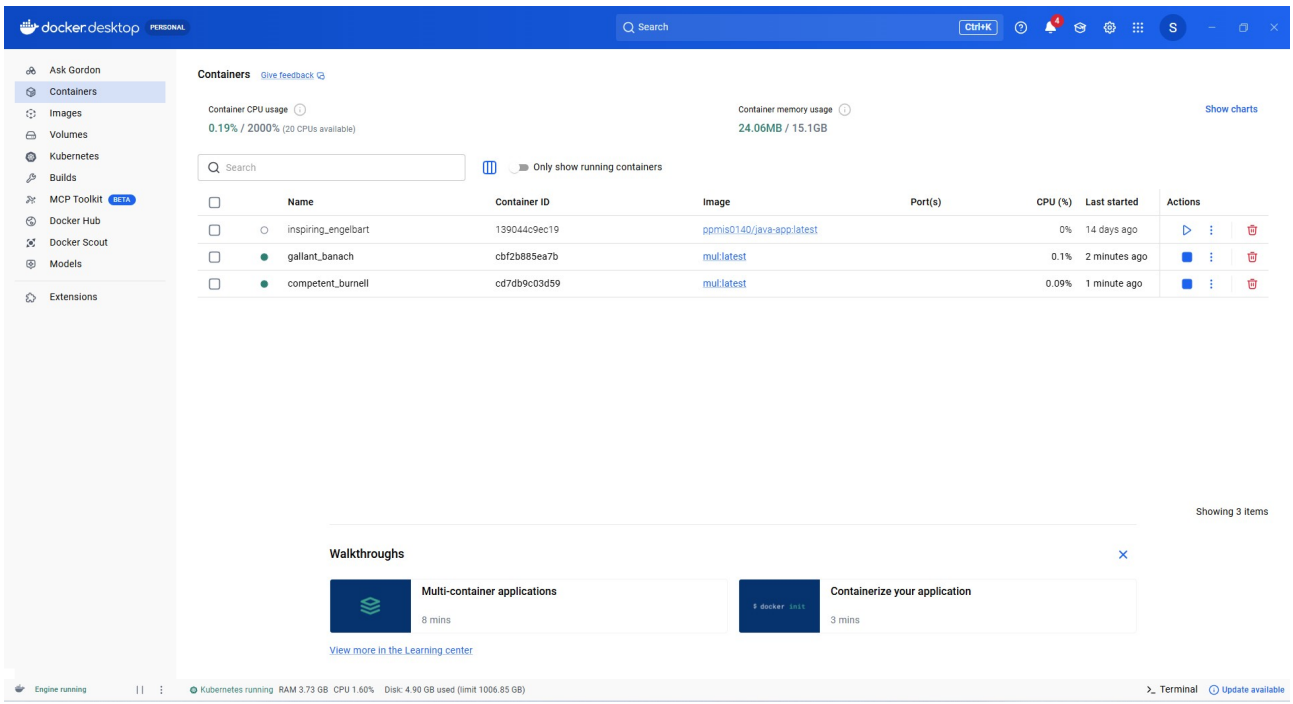


docker image:

name:mul



docker container


docker desktop PERSONAL
Search
Ctrl+K
Ask Gordon
Containers
Images
Volumes
Kubernetes
Builds
MCP Toolkit BETA
Docker Hub
Docker Scout
Models
Extensions
Containers
Give feedback
Container CPU usage
0.19% / 2000% (20 CPUs available)
Container memory usage
24.06MB / 15.1GB
Show charts
Search
Only show running containers
Name
Container ID
Image
Port(s)
CPU (%)
Last started
Actions
inspiring_engelbart 139044c9ec19 ppmis0140/java-app:latest 0% 14 days ago
gallant_banach cbf2b885ea7b mul:latest 0.1% 2 minutes ago
competent_burnell cd7db9c03d59 mul:latest 0.09% 1 minute ago
Showing 3 items
Walkthroughs
Multi-container applications
8 mins
Containerize your application
3 mins
View more in the Learning center
Engine running
Kubernetes running RAM 3.73 GB CPU 1.60% Disk: 4.90 GB used (limit 1006.85 GB)
Terminal
Update available